

Q1. Peak Element - List

Score: 10.00 TC: 1/1 passed

CODE

```
numbers = [1, 3, 20, 4, 1, 0]

for i in range(len(numbers)):
    if (i == 0 or numbers[i] ≥ numbers[i - 1]) and (i == len(numbers) - 1 or numbers[i] ≥ numbers[i + 1]):
        print(numbers[i])
        break
```

OUTPUT

20

Q2. Second Largest and Second Smallest - Tuple

Score: 0.00 TC: 0/1 passed

CODE

```
numbers = (10, 20, 5, 40, 15, 30)
sorted_tuple = tuple(sorted(numbers))

second_smallest = sorted_tuple[1]
second_largest = sorted_tuple[-2]

print(second_smallest)
print(second_largest)
```

OUTPUT

10
30**Q3. Frequency of Each Element - List**

Score: 10.00 TC: 1/1 passed

CODE

```
numbers = [1, 2, 3, 2, 4, 1, 5, 2]
freq = {}
for x in numbers:
    freq[x] = freq.get(x, 0) + 1
for key, value in freq.items():
    print(f"{key} = {value}")
```

OUTPUT

1 = 2
2 = 3
3 = 1
4 = 1
5 = 1**Q4. Missing Number - Tuple**

Score: 10.00 TC: 1/1 passed

CODE

```
numbers = (1, 2, 3, 5, 6)
n = max(numbers)
missing = set(range(1, n + 1)) - set(numbers)
print(missing.pop())
```

OUTPUT

4

Q5. Majority Element -List

Score: 10.00 TC: 1/1 passed

CODE

```
def find_majority_element(numbers):
    n = len(numbers)
    for num in set(numbers):
        if numbers.count(num) > n // 2:
            return num
    return None
numbers = [2, 2, 1, 2, 3, 2, 2]
result = find_majority_element(numbers)
if result is not None:
    print(result)
else:
    print("No majority element")
```

OUTPUT

2

Q6. Sum & Average - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
numbers = (10, 20, 30, 40, 50)
total = sum(numbers)
average = total / len(numbers)
print(total)
print(average)
```

OUTPUT150
30.0**Q7. Element Exists - Tuple**

Score: 10.00 TC: 2/2 passed

CODE

```
numbers = tuple(map(int, input().split()))
element = int(input())

if element in numbers:
    print("Element found")
else:
    print("Element not found")
```

INPUT10 20 30 40 50
30**OUTPUT**

Element found

Q8. Maximum Element - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
numbers = (12, 45, 8, 67, 23)
largest = max(numbers)
print(largest)
```

OUTPUT

67

7/31/26, 2:04 PM

Student Submissions — Class Work (30.7.26)

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Q9. Lucky Number

Score: 10.00 TC: 2/2 passed

CODE

```
def is_lucky(num):
    if num < 10:
        return num == 7
    digit_sum = sum(int(d) for d in str(num))
    return is_lucky(digit_sum)

number = int(input())

if is_lucky(number):
    print("Lucky Number")
else:
    print("Not a Lucky Number")
```

INPUT

34

OUTPUT

Lucky Number

Q10. Remove Duplicate Elements - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
numbers = (10, 20, 30, 20, 40, 10, 50)
unique_tuple = tuple({x for x in numbers[::-1]})
print(unique_tuple)
```

OUTPUT

(40, 10, 50, 20, 30)